Lecture 26: Introduction to Diodes

We now move to field of 'Electronics'.

Electrical Sciences: Flow of electronics in circuit with elements such as conductors, metals, semiconductor, vaccum etc and combination of these mediums

Electronics: Flow of electrons in medium like gas, vaccum, semiconductor comes under subject 'electronics'.

Conductors: When voltage is applied across a materials, of electrons start flowing freely ⇒ metals, silver, copper, iron, gold, aluminium

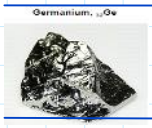
Insulators: When voltage is applied, electrons are not freely available to flows: Such materials are insulators such as plastic, paper wood, mica etc..

Semiconductors: When voltage is applied, number of electrons and their flow are achieved in controlled manner. These materials are called semiconductors, such as Silicon, Germanium etc

Conductivity wise: $\sigma_{insulator} < \sigma_{semi} < \sigma_{conductors}$

Thus, conductivity of Semiconductor material is controlled by applying voltage or electric field across the material.

Generic Semiconductor Materials

	Semiconductor Materials	Atomic No.	Valance electrons (Covalent bond), Tetravalent structure
	Silicon (Si)	14 = 2,8,4	4
	Germanium (Ge)	32 = 2,8,18,4	4

In pure Silicon or Germanium crystal Each atom in Si or Ge crystal, contributes 4 valence electrons, so the item is tetravalent. The atoms share covalent bond with nearest 4 atoms. The electron pair or covalent bond is represented by two dashed lines.

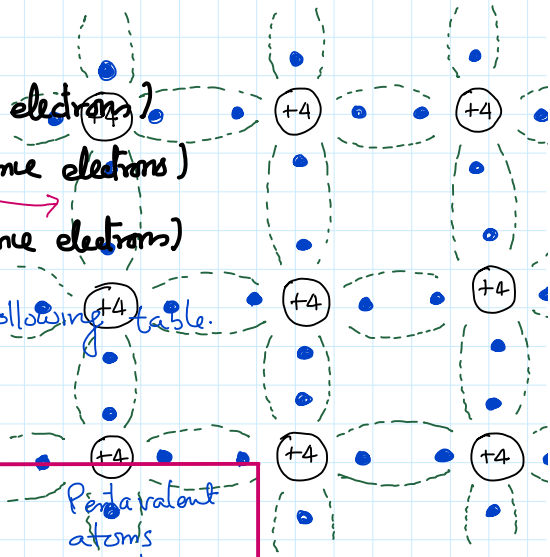
Valence electrons bind one atom to the next, at the same time valence electrons also tightly bound to the nucleus. This gives the low conductivity of these pure semiconductor materials.

tetrahedron

Types of Atoms based on the Valence Electrons:

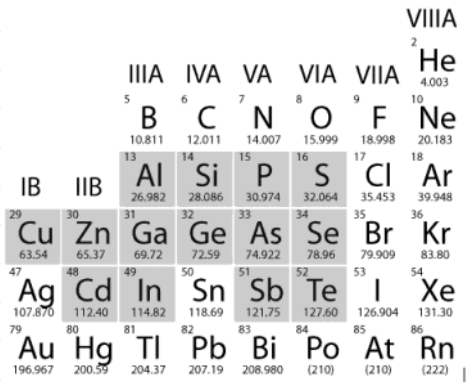
1. Trivalent (3 valence electrons)
2. Tetravalent (4 valence electrons)
3. Pentavalent (5 valence electrons)

These are given in the following table.



Valence Shell Electrons	Trivalent Atoms (III)	Tetravalent atoms (IV)	Pentavalent atoms (V)
	5 Boron B = 2,3	6 Carbon C = 2,4	7 Nitrogen N = 2,5
	13 Aluminium Al = 2,8,3	14 Silicon Si = 2,8,4	15 Phosphorus P = 2,8,5
	31 Gallium Ga = 2,8,18,3	32 Germanium Ge = 2,8,18,4	33 Arsenic As = 2,8,18,5
	49 Indium In = 2,8,18,18,3	50 Tin Sn = 2,8,18,18,4	51 Antimony Sb = 2,8,18,18,5

Silicon Germanium Crys



In = 2,8,18,18,3	Sn = 2,8,18,18,4	Sb = 2,8,18,18,5

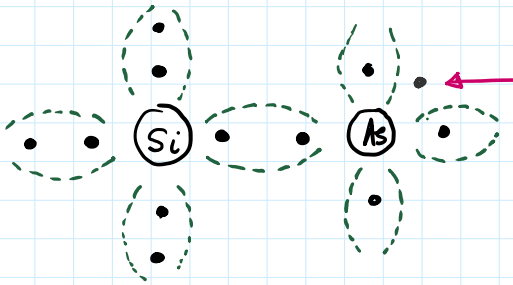
To increase conductivity of semiconductor materials impurities are added. These impurities are of two types.

1- n-type impurity ⇒ Pentavalent atoms such as Phosphorus $_{15}$ = 2,8,5 (valence electrons = 5)

Arsenic $_{33}$ = 2,8,18,5 (valence electrons = 5)

2. p-type impurity ⇒ Trivalent atoms such as Boron $_{5}$ = 2,3 } valence electron = 3

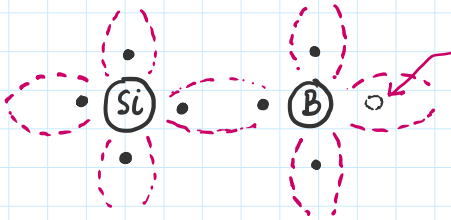
Indium $_{49}$ = 2,8,18,18,3 } valence electron = 3



n-type impurity

electrons are majority carriers

(electrons easier to free)

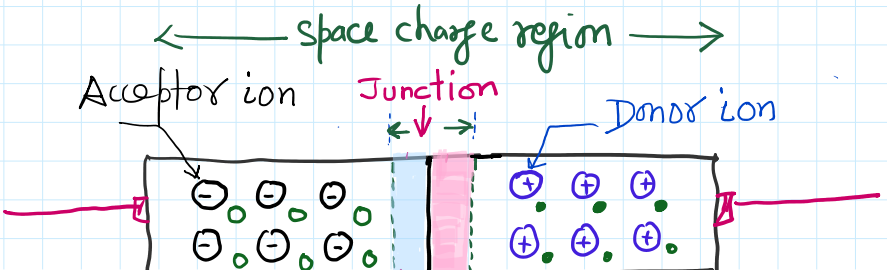


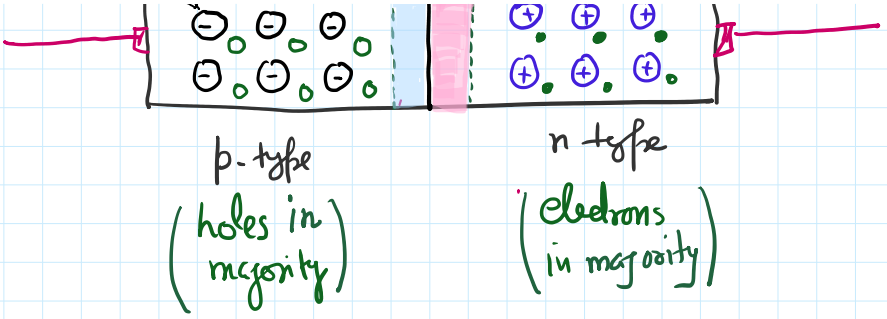
p-type impurity (absence of electron = hole) | holes are majority carriers

(holes easier to free)

PN Junction Diode:

When crystal lattice with n and p-type impurities are joined through a process, a junction is formed, termed as pn junction or junction diode. Obviously it is two termind device, one termind on each side of structure. This is shown below.





Space Charge Region:

Once the two types of semiconductor materials are joined through a process, there is density* gradient, due to majority of electrons on the n-side and majority of holes on the p-type structure.

As a natural phenomenon, the holes will initially diffuse to the right of the junction and electrons diffuse to the left side, due to high concentration and pull due to opposite charge present in the crystal.

When electrons pass through junction toward left, these get combined with holes crossing junction towards the right, or vise-versa.

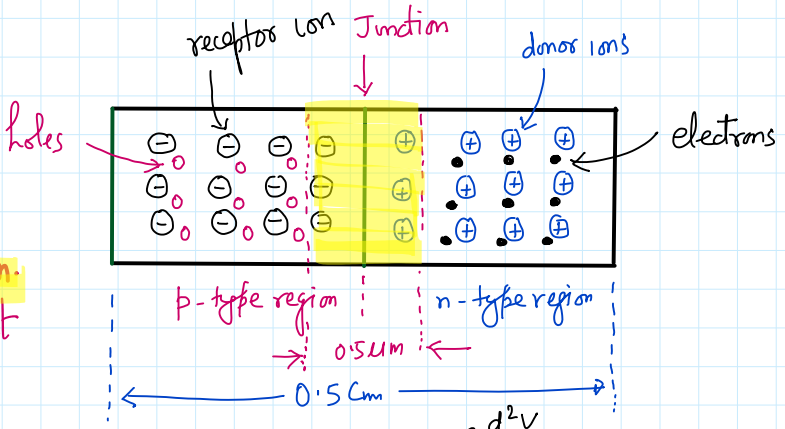
As a result of this, the unneutralized (charged) ions thus remain present near the junction. Thus negative ions are present on the left side of junction and positive side are present on the right side of the junction.

This forms the space charge region as shown below.

* Due to the concentration gradient, the electrons, holes flow forming a current called Diffusion Current. This happens initially, when a pn junction is formed.

This happens initially, when a pn junction is formed.

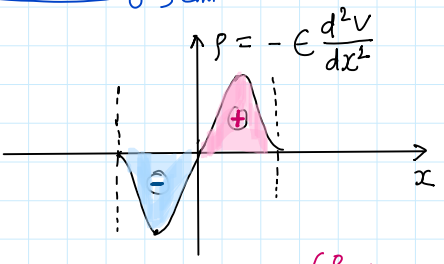
** Another current, which normally flows in pn junction is drift current. This results due to application of an electric field across the pn junction diode.



Poisson's Distribution eqn.
named after French Scientist Siméon Denis Poisson

$$\frac{d^2V}{dx^2} = -\frac{\rho}{\epsilon}$$

or $\rho(x) = -\epsilon \frac{d^2V}{dx^2}$



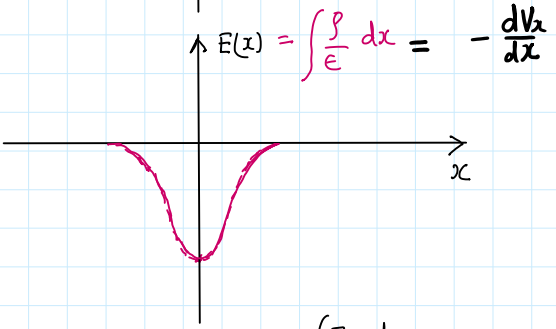
By definition:

$$E_x = -\frac{dV}{dx} = -\int\left(-\frac{\rho}{\epsilon}\right)dx$$

$$E_x = \int \frac{\rho}{\epsilon}\,dx$$

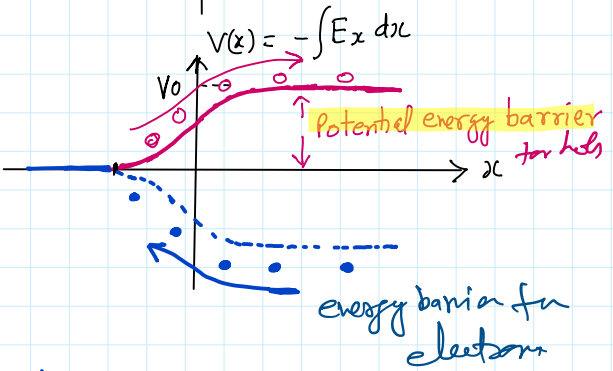
$$\downarrow$$

$$V_x = -\int E_x\,dx$$



Electrons on hill side has to climb up to overcome potential energy = eV

Holes in p-region has to climb the potential energy up = qV.



Thus p-n junction forms a potential barrier.
